PKU–Zurich PhD Summer School on Machine Learning for Macroeconomics and Finance

A baseline model of climate and the economy (DICE/CDICE), solved with Deep Equilibrium Nets

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Session 6 — the baseline climate–economy model, solved with DEQNs.

Tuned production code:

https://github.com/ClimateChangeEcon/Climate_in_Climate_Economics

https://liboecon.com/2026_summer_school.html

Where This Session Sits

This session (Day 2)
Baseline model of climate
and the economy:
DICE/CDICE + DEQN

→

Day 3 (Kübler–Scheidegger)
Climate *economics*:
heterogeneous agents,
constrained-optimal carbon taxes

Today we build and solve the baseline climate–economy model. The deeper climate *economics* — optimal carbon taxation with heterogeneous agents — follows on Day 3.

In this session:

1. Motivation: why an integrated assessment model (IAM)?
2. The DICE/CDICE model of climate and the economy
3. Solving the deterministic baseline with a Deep Equilibrium Net
4. A stochastic extension, and a look at deep uncertainty quantification

Learning objectives

By the end of this lecture you should be able to:

- Derive climate-economic coupling equations from Lagrangian FOCs and envelope conditions
- Implement deterministic and stochastic IAMs with DEQNs handling nonstationary exogenous sequences
- Construct GP surrogates for SCC and welfare outputs via Bayesian active learning
- Compute global sensitivity decompositions (Sobol, Shapley) to identify parameter drivers of uncertainty
- Apply deep-UQ pipelines to quantify tail distributions and risk in long-horizon climate projections

Part I

Why Climate Economics?

Climate Change as an Economic Threat

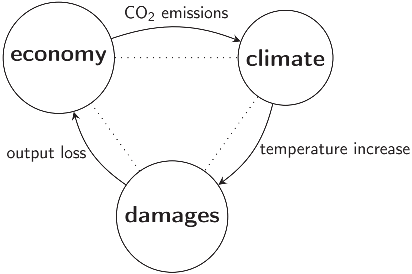
Observed global warming motivates the IAM feedback loop below.

- Global warming is a growing threat to economic well-being.
- Tipping points place us in catastrophic danger.
- Impacts current and future generations in different regions very differently.
- Recent overviews: Dietz (2024), van der Ploeg and Rezai (2026), Fernandez-Villaverde, Gillingham, Scheidegger (2025).

Why Integrated Assessment Models (IAMs)?

IAM logic

Couple the economy (output, investment, abatement, welfare) with the climate system (carbon concentrations, temperature, damages).



Output of interest: social cost of carbon (SCC), optimal tax paths, welfare decomposition.

The Social Cost of Carbon (SCC)

Definition

SCC at time t: marginal welfare cost of one extra unit of emissions, converted to consumption units.

$$\mathrm{SCC}_t = -\frac{\partial V_t / \partial E_t}{\partial V_t / \partial C_t}$$

High SCC when

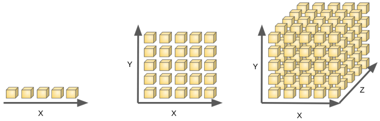
- Damages are steep
- Climate response is strong
- Discounting is low
- Tipping risk is material

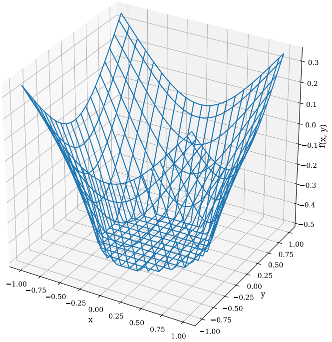
Policy interpretation

- Tax path for carbon pricing
- Benchmark for regulation / cap-and-trade
- Input to cost-benefit analysis
- Why uncertainty matters: SCC is a distribution

Why This Is Computationally Hard

- **Nonstationarity**: TFP, population, emissions intensity, and forcing paths all change exogenously over time.
- **Coupled dynamics**: economy and climate interact in both directions.
- **Long horizon**: welfare effects unfold over 100–300 years.
- **Curse of dimensionality**: nonlinearities, tipping hazards, constraints.





Our Contribution

- **Generic global solution method** to efficiently compute global solutions to stochastic IAMs that include economic and climate risks.
- **Perform global sensitivity analysis** with respect to uncertain parameters most discussed in the literature.

1. Exploit **deep equilibrium nets** (DEQNs) to alleviate the curse of dimensionality.
2. Add model parameters as **pseudo-states** $\rightarrow$ solve the model only once.
3. Construct **surrogate models** for the SCC (and other QoIs) using **Gaussian Process regression** and **Bayesian active learning**.

Part II

The DICE/CDICE Climate Model

DICE Baseline: Minimal Global IAM

State variables

- Economic stock: K_t
- Carbon stocks: $M_t^{\mathrm{AT}}, M_t^{\mathrm{UO}}, M_t^{\mathrm{LO}}$
- Temperature states: $T_t^{\mathrm{AT}}, T_t^{\mathrm{OC}}$
- Exogenous trends: population L_t, TFP A_t, emissions intensity σ_t

Controls

- Savings / investment share s_t
- Abatement rate μ_t

Why DICE is influential

- Transparent
- Fast to evaluate
- Policy-facing
- Benchmark for extensions

Challenge

Fast is useful, but only if the climate module is calibrated consistently with climate science.

Carbon Cycle (3-Box Model)

- DICE-2016 models the carbon cycle via three carbon reservoirs:
  - Atmosphere (AT), Upper ocean (UO), Lower ocean (LO)
- Aggregate distribution of carbon: $M_t = (M_{\mathrm{AT},t}, M_{\mathrm{UO},t}, M_{\mathrm{LO},t})^T$

Carbon concentrations evolve as a linear dynamic system:

$$M_{t+1} = (I + \Delta_t \cdot B) \cdot M_t + \Delta_t \cdot E_t$$

- Carbon emissions (economic activity + exogenous source): $E_t = E_{\mathrm{ind},t} + E_{\mathrm{Land}}$
- Industrial emissions: $E_{\mathrm{ind},t} = (1 - \mu_t)\sigma_t K_t^{\alpha}(A_t L_t)^{1-\alpha}$
- B is the transfer matrix governing exchange rates between reservoirs

PAUSE — 5 min Finger Exercise: One-Step Carbon Cycle

Exercise

The simplified carbon cycle has atmospheric CO_2 evolving as:

$$M_{\mathrm{AT}}^{t+1} = E_t + \phi_{11}\, M_{\mathrm{AT}}^{t} + \phi_{21}\, M_{\mathrm{UP}}^{t},$$

where $\phi_{11} = 0.912$, $\phi_{21} = 0.0444$, and M_{UP}^{t} (upper ocean) is given.

1. With $M_{\mathrm{AT}}^{2020} = 851$ GtC, $M_{\mathrm{UP}}^{2020} = 628$ GtC, and emissions $E_{2020} = 10$ GtC/yr (over a 5-year step: $E = 50$ GtC): compute M_{AT}^{2025}.
2. If emissions doubled to $E = 100$ GtC, what would M_{AT}^{2025} be?
3. By how many GtC does doubling emissions increase atmospheric CO_2?

Solution: One-Step Carbon Cycle

Answer

1. $M_{\text{AT}}^{2025} = 50 + 0.912 \times 851 + 0.0444 \times 628 = 50 + 776.1 + 27.9 = 854.0$ GtC.
2. $M_{\text{AT}}^{2025} = 100 + 776.1 + 27.9 = 904.0$ GtC.
3. Doubling emissions adds $904.0 - 854.0 = 50$ GtC—exactly the extra emissions. The carbon cycle is linear in emissions (at one-step horizon), so the effect is additive.

Temperature Dynamics (2-Box Model)

The two-layer energy balance model in DICE-2016:

$$
\begin{aligned}
T_{t+1}^{\mathrm{AT}} &= T_t^{\mathrm{AT}} + \Delta_t \cdot c_1 \left(F_t - \lambda T_t^{\mathrm{AT}} - c_3 \left(T_t^{\mathrm{AT}} - T_t^{\mathrm{OC}} \right) \right) \\
T_{t+1}^{\mathrm{OC}} &= T_t^{\mathrm{OC}} + \Delta_t \cdot c_4 \left(T_t^{\mathrm{AT}} - T_t^{\mathrm{OC}} \right)
\end{aligned}
$$

Radiative forcing:

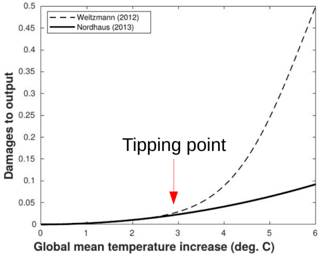
$$
F_t = F_{2\times\mathrm{CO}_2} \frac{\log(M_t^{\mathrm{AT}} / M_{\mathrm{PI}}^{\mathrm{AT}})}{\log(2)} + F_t^{\mathrm{EX}}
$$

Key parameters

$\lambda = F_{2\times\mathrm{CO}_2} / \Delta T_{\mathrm{AT},\times 2}$ where $\Delta T_{\mathrm{AT},\times 2}$ is the equilibrium climate sensitivity (ECS) — the long-run warming from CO_2 doubling. $M_{\mathrm{PI}}^{\mathrm{AT}}$ is the preindustrial atmospheric CO_2 stock.

Damage Functions

See also Tol (2009), Hänsel et al. (2020)



Damage fraction $\Omega(T)$: the share of gross output lost at warming T. Increasing in T.

Nordhaus (2008):

$$\Omega^N(T_{\mathrm{AT}}) = \pi_1 T_{\mathrm{AT}} + \pi_2 T_{\mathrm{AT}}^2$$

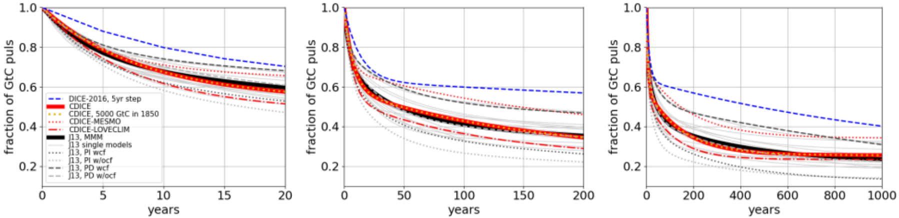
Weitzman (2012) — with tipping:

$$\Omega^W(T_{\mathrm{AT}}) = 1 - \frac{1}{1 + \left(\frac{1}{\psi_1} T_{\mathrm{AT}}\right)^2 + \left(\frac{1}{2\,TP} T_{\mathrm{AT}}\right)^{6.754}}$$

- Degree of convexity is of key importance in determining the optimal taxes [Weitzman 2012].
- Tipping points: losing the Amazon rain forest, faster onset of El Niño, reversal of the Gulf Stream, etc.

Why Recalibrate? → CDICE

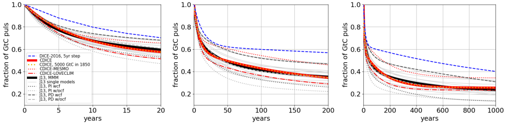
- Climate emulators must reproduce benchmark behavior from climate science model archives (CMIP).
- Earlier critiques highlighted major discrepancies for common IAM calibrations.
- Folini et al. show that the functional form can remain simple, but parameters require systematic calibration.



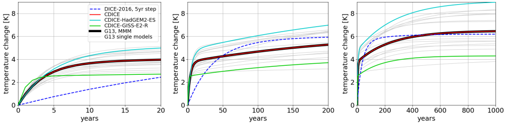
CDICE recalibrates the reduced-form climate emulator to climate-science benchmarks.

Four-Test Framework (Folini et al., 2025)

Test	Targets	Use
1. Carbon pulse (100 GtC)	Atmospheric retention path	Calibrate carbon cycle
2. $4\times CO_2$ step	Temperature impulse response	Calibrate temperature block
3. 1% CO_2/year	Transient climate response (TCR)	Out-of-sample validation
4. Historical + RCP	Realistic forcing paths	End-to-end validation



Carbon-pulse response.



Temperature response to a $4\times CO_2$ forcing step.

Equilibrium Climate Sensitivity (ECS) — A Key Uncertainty

IPCC AR5 gives a 'likely range' of [1.5K, 4.5K]

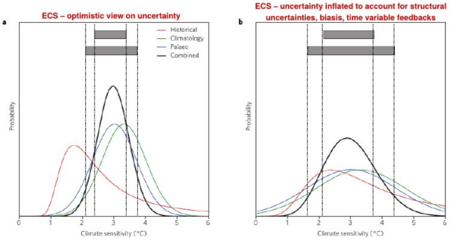


Figure 5 | Illustrative example of combining multiple constraints for climate sensitivity. Overall PDFs are the products of three PDFs based on the historical warming, climatological constraints on mostly GCMs, and palaeoclimate. Grey ranges at the top indicate a 'likely' (66%) and 'very likely' (90%) combined range. **a**,**b**, Constraint based on an optimistic interpretation of uncertainty ranges and assuming full independence (**a**) and on inflated ranges (**b**) to account for structural uncertainties, and with historical estimates inflated and scaled up to account for observation biases, and feedbacks varying from historical to future and across forcings. See Methods for details.

Knutti et al. 2017, Nature, doi: 10.1038/NGEO3017

- Roe and Baker (2007): $\mathrm{ECS} = \frac{\lambda_0}{1-f} F_{2\times\mathrm{CO}_2}$, $f \sim \mathcal{N}(\mu_f, S_f)$.
- Making ECS stochastic is a natural channel for Bayesian learning.

Transparent Time-Step Formulation

A practical contribution in CDICE: expose the time-step scaling directly in equations.

$$X_{t+\Delta t} = X_t + \Delta t \cdot f(X_t, u_t; \theta)$$

- Allows coherent annual, 5-year, or 10-year implementations within one generic formulation.
- Time discretization consistency improves transferability between classroom code, production code, and policy simulation horizons.

Summary of climate module:

Climate states (5)	$M_{\mathrm{AT}}, M_{\mathrm{UO}}, M_{\mathrm{LO}}, T^{\mathrm{AT}}, T^{\mathrm{OC}}$
Key parameters	ECS ($\Delta T_{\mathrm{AT},\times 2}$), carbon transfer B, forcing $F_{2\times \mathrm{CO}_2}$
Uncertainty channels	ECS, damage curvature, tipping

Part III

A Stochastic IAM — Full Derivation

The Planner's Problem — Epstein-Zin Preferences

The planner maximizes normalized recursive utility (Epstein-Zin):

$$V(\boldsymbol{X}_t)^{1-1/\psi} = \max_{C_t, K_{t+1}, \mu_t} \left\{ (1-\beta_t^{\mathrm{EZ}}) \left(\frac{C_t}{L_t}\right)^{1-1/\psi} L_t + \beta_t^{\mathrm{EZ}} \mathbb{E}_t \left[V(\boldsymbol{X}_{t+1})^{1-\gamma} \right]^{\frac{1-1/\psi}{1-\gamma}} \right\}$$

with $\beta_t^{\mathrm{EZ}} = \exp(-\rho\Delta_t)$.
subject to: budget constraint, capital accumulation, climate dynamics, learning dynamics.

Preference parameters

- ψ: intertemporal elasticity of substitution (IES)
- γ: risk aversion
- ρ: rate of time preference

Why Epstein-Zin?

Separate risk aversion from intertemporal substitution — crucial for long-run climate risk.

Full Constraint Set (10 Equations)

$$
\begin{aligned}
&\text{Budget:} && (1-\Omega(T_{\mathrm{AT},t})-\Theta(\mu_t))K_t^{\alpha}(A_tL_t)^{1-\alpha}-C_t-I_t=0\\
&\text{Capital:} && (1-\delta)K_t+I_t-K_{t+1}=0\\
&\text{Abatement bound:} && 1-\mu_t\geq 0\\
&\text{Carbon (AT):} && (1-\phi_{12})M_{\mathrm{AT},t}+\phi_{21}M_{\mathrm{UO},t}+(1-\mu_t)\sigma_tK_t^{\alpha}(A_tL_t)^{1-\alpha}+E_{\mathrm{Land},t}-M_{\mathrm{AT},t+1}=0\\
&\text{Carbon (UO):} && \phi_{12}M_{\mathrm{AT},t}+(1-\phi_{21}-\phi_{23})M_{\mathrm{UO},t}+\phi_{32}M_{\mathrm{LO},t}-M_{\mathrm{UO},t+1}=0\\
&\text{Carbon (LO):} && \phi_{23}M_{\mathrm{UO},t}+(1-\phi_{32})M_{\mathrm{LO},t}-M_{\mathrm{LO},t+1}=0\\
&\text{Temp (AT):} && T_{\mathrm{AT},t}+[\text{forcing}+\text{feedback}+\text{shock}]-T_{\mathrm{AT},t+1}=0\\
&\text{Temp (OC):} && \varphi_4T_{\mathrm{AT},t}+(1-\varphi_4)T_{\mathrm{OC},t}-T_{\mathrm{OC},t+1}=0\\
&\text{Learning }(\mu_f)\text{:} && \text{Bayesian posterior update}-\mu_{f,t+1}=0\\
&\text{Learning }(S_f)\text{:} && \text{Variance update}-S_{f,t+1}=0
\end{aligned}
$$

In this session's notebooks: 06_02/06_03 solve the 8-equation CDICE core (the first eight rows); the two Bayesian ECS-learning equations (μ_f, S_f) are the Friedl et al. research extension, not implemented here.

Nonstationarity and Time as a State Variable

The key difference from the discrete-time lectures

IAMs are nonstationary: A_t (TFP), L_t (population), σ_t (carbon intensity), $E_{\mathrm{Land},t}$, F_t^{EX} all change exogenously over time.

- Unlike Brock-Mirman or stationary OLG, we cannot have a time-invariant policy function.
- **Solution:** add t as an explicit state variable to the neural network input.
- For neural net stability, transform: $\tau = 1 - \exp(-\vartheta \cdot t) \in [0, 1)$
- Normalize capital, consumption, investment by $A_t \times L_t$ to detrend what we can.
- Time captures what remains nonstationary after detrending.

Practical implication

The policy network is $\mathcal{N}_\rho(\tau_t, \text{states}, \text{params})$ — transformed time is an input, not just an index.

State Space Summary

$$\boldsymbol{X}_t = \Big[\underbrace{k_t, M_{\mathrm{AT},t}, M_{\mathrm{UO},t}, M_{\mathrm{LO},t}, T_t^{\mathrm{AT}}, T_t^{\mathrm{OC}}, \mu_{f,t}, S_{f,t}, \tau_t}_{\text{9 endogenous/exogenous states}}; \underbrace{\theta}_{N \text{ pseudo-state parameters}} \Big]$$

Minimal model has a 9+N-dimensional state space.

Pseudo-state parameters θ

- ρ (time preference)
- γ (risk aversion)
- ψ (IES)
- π_2 (damage curvature)
- $\mu_{f,0}, S_{f,0}$ (ECS prior)
- Carbon cycle parameters

Why pseudo-states?

- Solve the model once for all parameter combinations
- UQ becomes interpolation on the learned surrogate
- From earlier DEQN lectures: the same trick used for β, α in Brock-Mirman

Derivation: The Lagrangian

Write the Lagrangian with multipliers (λ for budget, ξ for capital, η for abatement):

$$\begin{aligned}
\mathcal{L} = V(\boldsymbol{X}_t)^{1-1/\psi} \\
&+ \lambda_t \Big[\big(1 - \Omega(T_{\mathrm{AT},t}) - \Theta(\mu_t)\big) K_t^{\alpha} (A_t L_t)^{1-\alpha} - C_t - I_t \Big] \\
&+ \xi_t \Big[(1-\delta) K_t + I_t - K_{t+1} \Big] \\
&+ \eta_t \Big[1 - \mu_t \Big] \\
&+ [\text{multipliers for carbon cycle, temperature, learning dynamics}]
\end{aligned}$$

Control variables: C_t, K_{t+1}, μ_t

Strategy: Take FOCs $\rightarrow$ use envelope theorem $\rightarrow$ derive Euler equations

Derivation: First-Order Conditions

w.r.t. C_t (marginal utility = shadow price of budget):

$$\frac{\partial V^{1-1/\psi}}{\partial C_t} = \lambda_t$$

w.r.t. K_{t+1} (intertemporal trade-off):

$$\xi_t = e^{-\rho} \frac{\partial}{\partial K_{t+1}} \mathbb{E}_t \left[V(\boldsymbol{X}_{t+1})^{1-\gamma} \right]^{\frac{1-1/\psi}{1-\gamma}}$$

w.r.t. μ_t (marginal abatement cost = marginal emission damage):

$$\lambda_t \frac{\partial \Theta}{\partial \mu_t} K_t^{\alpha} (A_t L_t)^{1-\alpha} = \eta_t + [\text{shadow value of reduced emissions}]$$

Problem

Cannot compute $\partial V / \partial(\text{state}_{t+1})$ analytically $\rightarrow$ need the envelope theorem.

Derivation: Envelope Theorem

The envelope theorem gives us derivatives of the value function w.r.t. current states:

$$\frac{\partial V(\boldsymbol{X}_t)^{1-1/\psi}}{\partial k_t} = \lambda_t \cdot \alpha\big(1 - \Omega(T_{\mathrm{AT},t}) - \Theta(\mu_t)\big) k_t^{\alpha-1}(A_t L_t)^{1-\alpha} + \xi_t \cdot (1-\delta)$$

$$\frac{\partial V(\boldsymbol{X}_t)^{1-1/\psi}}{\partial M_{\mathrm{AT},t}} = [\text{shadow value of atmospheric carbon}]$$

$$\frac{\partial V(\boldsymbol{X}_t)^{1-1/\psi}}{\partial T_t^{\mathrm{AT}}} = [\text{shadow value of temperature}]$$

Strategy:

1. Get $\partial V/\partial(\text{state}_t)$ from envelope
2. Shift forward one period: $\partial V/\partial(\text{state}_{t+1})$
3. Substitute back into FOCs to obtain Euler equations

Derivation: Euler Equations

Capital Euler equation:

$$1 = e^{-\rho}\,\mathbb{E}_t\left[\left(\frac{V_{t+1}}{(\mathbb{E}_t[V_{t+1}^{1-\gamma}])^{1/(1-\gamma)}}\right)^{1/\psi-\gamma} \cdot \frac{(C_{t+1}/L_{t+1})^{-1/\psi}}{(C_t/L_t)^{-1/\psi}} \cdot R^K_{t+1}\right]$$

where R^K_{t+1} is the return on capital including climate damages.

SCC as shadow price of atmospheric carbon:

$$\mathrm{SCC}_t = -\frac{\partial V_t/\partial M_{\mathrm{AT},t}}{\partial V_t/\partial C_t}$$

Captures the present-value welfare cost of one extra unit of atmospheric carbon.

Abatement optimality

At the optimum: marginal abatement cost = carbon tax = SCC (in the planner's problem).

Bayesian Learning Mechanics (literature)

Temperature evolves with uncertain climate feedback $\tilde{f}_{t+1} \sim \mathcal{N}(\mu_{f,t}, S_{f,t})$:

$$T_{t+1}^{\mathrm{AT}} = \ldots + \varphi_{1C} \tilde{f}_{t+1} T_t^{\mathrm{AT}} + \tilde{\epsilon}_{T,t+1}$$

Gaussian conjugate posterior updates:

$$\mu_{f,t+1} = \frac{S_{\epsilon_T} \mu_{f,t} + \varphi_{1C} T_t^{\mathrm{AT}} \left(\varphi_{1C} T_t^{\mathrm{AT}} \tilde{f}_{t+1} + \tilde{\epsilon}_{T,t+1} \right) S_{f,t}}{S_{\epsilon_T} + \left(\varphi_{1C} T_t^{\mathrm{AT}} \right)^2 S_{f,t}}$$

$$S_{f,t+1} = \frac{S_{\epsilon_T} \cdot S_{f,t}}{S_{\epsilon_T} + \left(\varphi_{1C} T_t^{\mathrm{AT}} \right)^2 S_{f,t}}$$

- Agent observes shock realizations at the beginning of each period.
- Learning shifts optimal near-term abatement: insurance motive today vs. information value tomorrow.

From Euler Equations → DEQN Loss

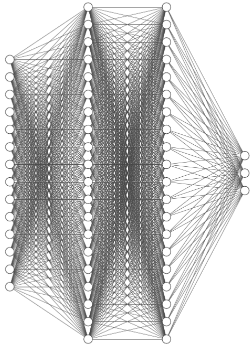
Each equilibrium condition becomes a residual:

$$G_i(\boldsymbol{x}, \mathcal{N}_{\boldsymbol{\rho}}) = 0 \quad \forall i$$

Loss function:

$$\ell_{\boldsymbol{\rho}} = \frac{1}{N} \sum_{\boldsymbol{x}_t} \sum_{i} G_i(\boldsymbol{x}_t, \mathcal{N}_{\boldsymbol{\rho}})^2$$

- No labeled data needed
- $\mathcal{N}_{\boldsymbol{\rho}}$ simulates the path itself
- Train with SGD (Adam)

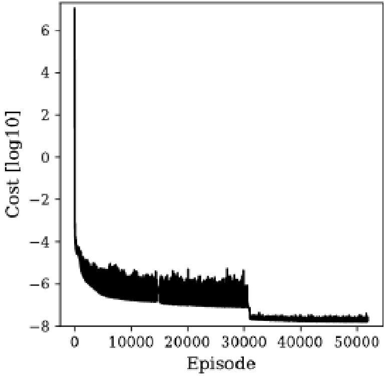


DEQN Architecture for the Stochastic IAM

Component	Specification
Hidden layers	2
Nodes per layer	512 (1024 in production)
Activation	relu
Optimizer	Adam, $\alpha = 5 \times 10^{-5}$
Batch size	512
Inputs	$9 + N$ (states + pseudo)
Outputs	k', μ, multipliers

Training workflow:

1. Simulate path with $\mathcal{N}_\rho$
2. Evaluate residuals on path
3. Gradient descent on loss
4. Mix broad & endogenous sampling



Training loss convergence

Epstein–Zin + i.i.d. weather (literature): model ingredients

Literature / preview — not a notebook in this session

The Epstein–Zin + i.i.d.-weather formulation in the next four slides follows Friedl et al. (2023) and is shown as an advanced target. This session's stochastic notebook (06_03) implements the simpler CRRA + AR(1) version (Cai–Lontzek spirit); EZ with tipping risk is developed further in the full course.

State (8-D): $\boldsymbol{X}_t = [k_t, M_{\mathrm{AT},t}, M_{\mathrm{UO},t}, M_{\mathrm{LO},t}, T_t^{\mathrm{AT}}, T_t^{\mathrm{OC}}, \tau_t, w_t]$

- $\tau_t = 1 - e^{-\vartheta t}$: time transform (handles nonstationarity).
- $w_t \sim \mathcal{N}(0, \sigma_w^2)$, iid: weather shock entering productivity and emissions intensity.

Network (single): $\mathcal{N}_\rho : \boldsymbol{X}_t \mapsto (k', \hat{\lambda}, \mu, \nu^{AT}, \nu^{UO}, \nu^{LO}, \eta^{AT}, \eta^{OC}, V) \in \mathbb{R}^9$.
Calibration: $\psi = 0.69$ (IES), $\gamma = 10$ (RA), $\sigma_w = 0.01$, 5-node Gauss–Hermite over w'.

Why a single 9-output network (Friedl style)?

Shadow values ν, η are network outputs, not derivatives of V. This gives the optimizer direct gradient signal on every multiplier instead of forcing it through second derivatives of V — crucial when V is itself being learned.

Epstein–Zin (literature): the 9-residual loss

With $\hat\beta = e^{-\rho}$, $e_1 = \frac{1-\gamma}{1-1/\psi}$, $e_2 = \frac{\gamma-1/\psi}{1-\gamma}$, $e_3 = \frac{1/\psi-\gamma}{1-1/\psi}$:

Capital Euler (eq 1):

$$e^{(g^A+g^L)\Delta t}\,\hat\lambda_t \;=\; \hat\beta\left(\mathbb{E}_t[(V_+/V)^{e_1}]\right)^{e_2}\mathbb{E}_t[(V_+/V)^{e_3}\,v_{k,+}]$$

Budget (eq 2), KKT-μ (eq 3, Fischer–Burmeister, no expectation).
Envelope conditions on $T^{AT}, M^{AT}, M^{UO}, M^{LO}, T^{OC}$ (eq 4–8):

$$\eta_t^{AT} = \hat\beta\,(\mathbb{E}_t[(V_+/V)^{e_1}])^{e_2}\,\mathbb{E}_t[(V_+/V)^{e_3}\,v_{T^{AT},+}], \quad \text{etc.}$$

EZ Bellman (eq 9, Bansal–Yaron normalized):

$$V_t^{1-1/\psi} = (1-\hat\beta)\,c_t^{1-1/\psi} + \hat\beta\left(\underbrace{\mathbb{E}_t[V_+^{1-\gamma}]}_{\textbf{not frozen}}\right)^{\frac{1-1/\psi}{1-\gamma}}$$

Loss: $\mathcal{L}(\theta) = \frac{1}{9}\sum_{i=1}^{9}\mathbb{E}_{\boldsymbol{X}\sim\pi_\theta}[G_i(\boldsymbol{X};\theta)^2]$ (equal-weighted MSE).

v_{t+1}/v_t normalization

We never evaluate $V_+^{e_3}$ directly ($e_3 \approx 19$ at $\gamma = 10$ blows up float32). Form ratios V_+/V first, then take the power — numerator and denominator stay $\mathcal{O}(1)$.

Freezing trick (literature): `stop_gradient` on EZ corrections

Pathology. At $\gamma = 10$, $\psi = 0.69 \Rightarrow e_3 \approx 19$. A 1% spread in V_+ across GH nodes becomes a 19% spread in $(V_+/V)^{e_3}$. The optimizer "discovers" it can shrink the FOC residual by moving V — but V is also pinned by the Bellman equation (eq 9). The two forces fight, training oscillates.
Fix. Wrap every EZ correction in `tf.stop_gradient`:

```
ez_corr = tf.stop_gradient(tf.pow(V_n / V, e3))
```

Forward pass keeps $\Phi = (V_+/V)^{e_3}$; backward pass treats Φ as a per-state constant.
Fixed-point view. Each Adam step is one iteration of:

1. Evaluate Φ at current θ *(frozen for backprop)*.
2. FOC residuals (eq 1, 4–8) train μ, λ, ν, η against frozen Φ.
3. Bellman residual (eq 9) trains V alone via $\mathbb{E}[V_+^{1-\gamma}]$ (un-frozen).

The fixed point is the joint EZ equilibrium — but the iteration is far better conditioned than full joint-gradient minimization at $\gamma = 10$.

One-line summary

Freeze high-power V-corrections in the FOCs; train V only via the Bellman equation. This stabilizes the high-γ training run.

γ-curriculum and diagnostics (literature)

Curriculum (200 episodes in the reference run):

Phase	Eps	γ
1 (CRRA basin)	0–60	$1/\psi \approx 1.45$
2 (linear ramp)	60–130	$1.45 \to 10$
3 (refine)	130–200	10

At $\gamma = 1/\psi$ all EZ exponents collapse ($e_2 = e_3 = 0$, corrections $\equiv 1$): phase 1 trains a clean CRRA solution; phase 2 brings the high-power corrections online slowly; phase 3 refines.

What we report (see Validation slide):

- Aggregate loss curve with γ-schedule overlay.
- Per-residual MSE for each of the 9 equations (every 5 eps).
- Final residual statistics: mean / RMSE / median / p95 / max.
- SCC distribution 2015–2100 from 100-path Monte Carlo.

Economic punchline

Symmetric iid Gaussian weather $\Rightarrow$ EZ precautionary tilt on SCC is small (a few %). The Cai–Lontzek "EZ triples SCC" result needs *skewed* risk (tipping points) — see the OLG-IAM with stochastic tipping in the full course.

Validation

- Track residual quantiles, not only mean loss.
- Simulate long paths and inspect feasibility violations.
- Stress-test under extreme carbon and temperature states.
- Compare policy moments against known benchmarks.
- Verify robustness to seeds and learning-rate schedules.

Minimum reporting standard

Always report Euler/residual distributions, not only selected trajectories.

Part IV

Deep Uncertainty Quantification

Why Global UQ?

Brute-force UQ workflow

1. Draw parameter vector $\theta^{(m)}$
2. Solve full stochastic IAM
3. Simulate, compute SCC and welfare
4. Repeat for thousands of draws

Problem

If one model solution is expensive, global sensitivity analysis becomes infeasible.

Our approach

- One-time expensive training with pseudo-states
- Then cheap evaluation everywhere in parameter space
- "One-at-a-time" sensitivity is local and misleading
- We need global sensitivity analysis

The Pseudo-States Trick

Idea

Treat uncertain parameters as additional state inputs to one global policy network.

$$\tilde{\boldsymbol{x}}_t = (\underbrace{\boldsymbol{x}_t}_{\text{econ + climate states}}, \underbrace{\theta}_{\text{uncertain params}}), \qquad u_t = \mathcal{N}_\rho(\tilde{\boldsymbol{x}}_t)$$

$$\text{State } X_t = (k_t, M_t^{AT}, M_t^{UO}, M_t^{LO}, T_t^{AT}, T_t^{OC}, \zeta_t, \chi_t, TP_t, t; \theta)$$

$$\text{Policy } p_t = (k_{t+1}, c_t, \mu_t)$$

One model training approximates policy functions for all parameter values simultaneously.

GP Surrogates for Quantities of Interest

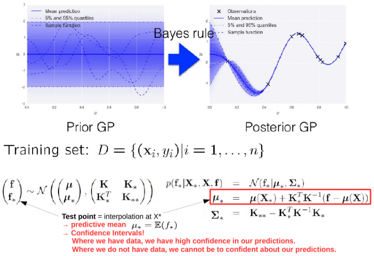
Construct training data:

1. Sample parameter vectors $\theta^{(m)}$
2. Simulate model with DEQN surrogate
3. Compute QoIs: SCC, welfare, moments

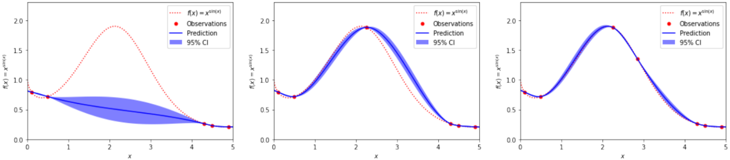
Fit Gaussian process:

$$q(\theta) \sim \mathcal{GP}(m(\theta), k(\theta, \theta'))$$

- Predictive mean: fast approximation
- Predictive variance: uncertainty-aware



Bayesian Active Learning (BAL)



- Explore where GP uncertainty is high; exploit where objective looks promising.
- Add new expensive simulations only where needed.
- Far fewer expensive model evaluations to reach stable UQ conclusions.

Global Sensitivity Metrics: Sobol Indices and Shapley Effects

Sobol first-order index:

$$S_i = \frac{\mathrm{Var}\left(\mathbb{E}[q(\theta) \mid \theta_i]\right)}{\mathrm{Var}(q(\theta))}$$

Total-effect index:

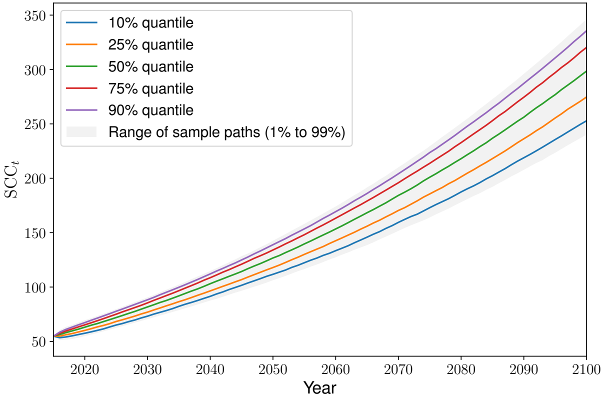
$$S_i^{\mathrm{tot}} = 1 - \frac{\mathrm{Var}\left(\mathbb{E}[q(\theta) \mid \theta_{-i}]\right)}{\mathrm{Var}(q(\theta))}$$

- Sobol: variance decomposition perspective
- Shapley: fair attribution including interactions
- Together: robust ranking of drivers of SCC dispersion

Result

Typically: ECS uncertainty and damage specification dominate SCC variability.

Results: SCC Distributions



- SCC realizations are wide and right-skewed.
- Learning shifts mass away from extreme tails over time.
- Policy design should target tail insurance, not only mean outcomes.

Key Takeaways from This Session

1. **Climate calibration matters**: CDICE-style tests anchor the climate block in science benchmarks.
2. **DEQNs scale nonstationary IAMs**: full Lagrangian $\rightarrow$ FOCs $\rightarrow$ envelope $\rightarrow$ Euler $\rightarrow$ residual loss.
3. **Time as state variable**: the critical bridge from stationary models (earlier discrete-time lectures) to real-world applications.
4. **Pseudo-states enable one-shot UQ**: solve once, evaluate everywhere via GP + BAL.

Next

Day 3 (Kübler–Scheidegger): extend from representative-agent IAM to OLG with heterogeneous agents $\rightarrow$ constrained Pareto-improving carbon tax rules.

Notebook Roadmap for This Session

The companion notebooks for this lecture sit in `code/06_climate/` and are intended to be opened in this order:

1. `06_01_Climate_Exercise.ipynb` – pure NumPy warm-up. Three-box carbon cycle, single-layer temperature, quadratic damages, BAU vs 50% abatement. No neural networks: builds intuition for the climate module before the DEQN.
2. `06_02_DICE_DEQN_Library_Port.ipynb` – deterministic CDICE solved with DEQN. Self-contained port of the production `DEQN_for_IAMs/dice_generic` library, with the Friedl τ-transformation and a verification gate against the reference solution.
3. `06_03_Stochastic_DICE_DEQN.ipynb` – (preview / self-study extension) AR(1) productivity shock, Gauss–Hermite quadrature on the Euler expectation, Monte-Carlo SCC fan charts in the Cai–Lontzek spirit.

References and Reading Guide

Core references for this lecture

- Nordhaus (2017): DICE architecture and SCC framework.
- Folini, Friedl, Kübler, Scheidegger (2025, *Restud*): CDICE calibration and tests.
- Friedl, Kübler, Scheidegger, Usui (2023): deep UQ for stochastic IAMs.
- Dietz (2024), van der Ploeg and Rezai (2026), Hassler, Krusell, Smith (2016): climate-economics and IAM reviews.
- Azinovic, Gaegauf, Scheidegger (2022): DEQN methodology.

Questions?